

Student Dropout Risk Prediction (Final Project Report)

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Bootcamp Cohort: Tech4Dev WT Bootcamp Cohort 5.1

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1. Executive Summary

Student dropout is a persistent challenge in higher education, affecting students' academic success, institutional performance, and long-term career opportunities. Early identification of at-risk students enables institutions to provide timely academic and welfare support, improving retention and overall student outcomes.

This project builds a data-driven Early Warning System to help academic advisors identify at-risk students before they withdraw. Using a 9,500-record student dataset, the team analyzed behavioral, academic, and demographic predictors of dropout, built a classification model to score dropout likelihood, and designed a risk dashboard with tiered (Low/Medium/High) outputs alongside a non-punitive advisor intervention guide.

The solution demonstrates how educational data can support evidence-based decision-making by enabling earlier interventions and more efficient allocation of institutional support resources.

2. Project Background & Objective

2.1 Problem Statement

Many students who eventually withdraw show measurable warning signs, such as declining grades, poor attendance, and rising stress, well before they leave, yet institutions rarely have a systematic way to detect these signs in time to intervene. This project closes that gap by identifying the strongest predictors of dropout and surfacing at-risk students early enough for advisors to take meaningful, non-punitive action.

The project supports SDG 4 (Quality Education) by aiming to improve completion rates, and SDG 10 (Reduced Inequalities) by highlighting how socio-economic and support factors disproportionately drive dropout risk among disadvantaged students.

Beyond improving student outcomes, the proposed system can also help higher education institutions prioritize limited academic support resources by identifying students who require early intervention.

2.2 Objectives

- Analyze the student dataset to identify the strongest behavioral, academic, and demographic predictors of dropout.
- To identify the factors associated with student dropout.
- Design a risk dashboard presenting each student's risk score, key contributing factors, and risk tier (Low/Medium/High).
- Deliver a documented, demo-ready solution by the end of project week, with transparent AI tool usage disclosure.

2.3 Scope

In scope: Data cleaning, exploratory data analysis EDA on the provided dataset; a dropout-risk classification model with feature-importance outputs; a risk dashboard (score, contributing factors, tier); a non-punitive intervention-pointer guide per risk; and documentation of methodology, AI usage, and the final Demo Day presentation.

Out of scope: integration with any live institutional student information system (SIS) or production deployment; real-time data pipelines or automated refresh beyond the static dataset; direct outreach, counselling, or intervention delivery to students (recommendations only); and long-term model monitoring, retraining, or maintenance beyond the capstone period.

2.4 Project Execution Summary

The project was executed using a one-week sprint approach, with clearly defined milestones, deliverables, and responsibilities outlined in the Project Charter. The Technical Project Managers coordinated project planning, monitored progress against the schedule, facilitated communication among

team members, tracked project risks, and ensured that deliverables met the agreed Definition of Done and Acceptance Criteria.

Regular team check-ins and progress reviews were conducted throughout the project to identify blockers early, monitor task completion, and maintain alignment between project management and data analysis activities. This structured approach enabled the team to complete the project within the scheduled timeline despite the short development window.

2.5 Team Collaboration

The project adopted a collaborative approach, with responsibilities assigned according to each member's area of expertise. Technical Project Managers led project planning, documentation, scheduling, and risk management, while the Data Analysts focused on data preparation, exploratory analysis, feature engineering, predictive modelling, dashboard development, and validation.

Shared documentation, continuous communication, and regular feedback sessions ensured consistency across all project activities. This collaborative workflow improved coordination, encouraged knowledge sharing, and supported timely delivery of all project milestones.

3. Methodology

3.1 Data Overview

The dataset initially contained **10,001 student records** across **19 variables**, including both numerical and categorical features. After removing records with missing values, the final cleaned dataset comprised **9,501 records** and **19 columns**, providing a reliable foundation for further analysis.

The dataset included **9 numerical variables** (such as age, study hours, attendance rate, exam score, family income, stress level, and travel time) and **10 categorical variables** (including gender, internet access,

scholarship status, parental education, department, semester, and dropout status).

Descriptive statistical analysis was performed on all numerical variables using measures such as **mean, median, standard deviation, minimum, and maximum values**. Frequency analysis was conducted for categorical variables, revealing:

- **Gender:** Evenly distributed (50% female, 50% male)
- **Internet Access:** 88% of students had internet access, while 12% did not.
- **Parental Education:** Bachelor's degree (39.6%) was the most common educational level, followed by High School (30.6%), Master's (20.0%), None (5.1%), and PhD (4.7%).
- **Dropout Rate:** Approximately **23.6% (2,241 students)** had dropped out, providing a sufficiently balanced target variable for predictive modeling.

Comparative analysis between students who dropped out and those who remained enrolled identified several key behavioral differences:

Metric	Non-Dropout	Dropout	Observation
Attendance Rate	~82.7%	~79.5%	Lower attendance among dropouts
Stress Level	~5.26	~6.32	Higher stress among dropouts
Study Hours per Day	~4.08	~3.80	Dropouts studied fewer hours
Assignment Delay	~1.74 days	~2.00 days	Dropouts submitted assignments later
Family Income	~\$38.5K	~\$38.0K	Minimal difference

Key Insights

- **Stress level** emerged as the strongest behavioral differentiator, with dropouts reporting approximately **20% higher stress** than non-dropouts.
- **Attendance rate, study hours, and assignment submission delays** also showed meaningful differences, indicating these factors are likely strong predictors of student retention.
- **Family income** showed little variation between the two groups, suggesting it is **not a strong standalone predictor** of dropout and may require interaction analysis with variables such as part-time employment or scholarship status.
- **Gender differences were negligible**, as dropout patterns were consistent across male and female students, reducing concerns about gender bias in subsequent predictive models.

Tools Used: Google Sheets for data cleaning and exploratory analysis, with Claude AI providing analytical support and interpretation.

3.2 Data Cleaning

Of the **10,000 records** in the dataset, **980 data points across 500 rows** were null. The missing data was limited to **four variables: Parental Education (500 missing values), Family Income (500), Study Hours per Day (500), and Stress Index (500)**. Removing all incomplete records reduced the dataset to **9,500 rows observations**.

After removing records with missing values, the final cleaned dataset comprised **9,500 records** and **19 columns**, providing a reliable foundation for further analysis.

3.3 Exploratory Data Analysis — Key Findings

Comparing students who dropped out versus those who remained enrolled revealed consistent patterns across attendance, study habits, stress levels, and assignment completion.

Comparative analysis between students who dropped out and those who remained enrolled identified several key behavioral differences:

Metric	Non-Dropout (0)	Dropout (1)
Attendance Rate (%)	82.7 (avg. across genders)	79.5 (avg. across genders)
Study Hours/Day	4.08	3.80
Stress Index	5.26	6.31
Assignment Delay (days)	1.74	2.00
Group Size	7,259 (76.41%)	2,241 (23.59%)

Students who dropped out generally demonstrated:

- Lower attendance rates (~79.5% vs. ~82.7%).
- Fewer study hours per day (~3.80 vs. ~4.08).
- Higher stress levels (~6.31 vs. ~5.26).
- Longer assignment delays (~2.00 vs. ~1.74 days).

These findings suggest that dropout is influenced by a combination of academic performance, learning behaviour, and student wellbeing rather than a single factor. Students exhibiting multiple warning indicators are therefore more likely to benefit from timely academic support before withdrawal becomes likely.

The variables above were selected as the primary candidate features for feature engineering and predictive modelling.

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3.4 PREDICTIVE Modeling

Key Findings Summary — Student Dropout Analysis

Dataset: 9,500 students | **Dropout rate:** 23.6% (2,241 students)

Factor	Finding
Stress Index	Strongest link to dropout. Dropout students average stress ~6.3 vs ~5.3 for those who stayed — true for both genders.
Internet Access	No access → 28.4% dropout vs 22.9% with access. Clear gap.
Part-Time Job	Job holders drop out more (25.6%) vs no job (22.2%).
Scholarship	Small protective effect: 23.3% (with) vs 23.8% (without).
Gender	No real difference (Female 23.8% vs Male 23.4%).
Department	Even across all 5 departments (~23–24%). No standout.
Parental Education	Slight downward trend with higher education, but weak overall.
Travel Time	Dropout group slightly higher (30.85 vs 29.95 min) — minor factor.
Attendance & Study Hours	Lower attendance/study time trends toward higher dropout risk.
Assignment Delays	Frequent delays skew toward the dropout group.

Bottom line: Dropout is driven mainly by **stress, lack of internet access, and part-time work**. Demographic factors (gender, department, parental background) show little to no effect on their own.

4. Risk Tiering & Dashboard

The Early Warning Dashboard categorises students into **Low**, **Medium**, and **High** dropout-risk tiers using model-generated probability scores.

The dashboard enables academic advisors to quickly identify students requiring immediate attention while also displaying the major factors contributing to each student's risk score.

This tiered approach allows institutions to prioritise interventions based on risk severity, ensuring available support resources are directed where they are likely to have the greatest impact.

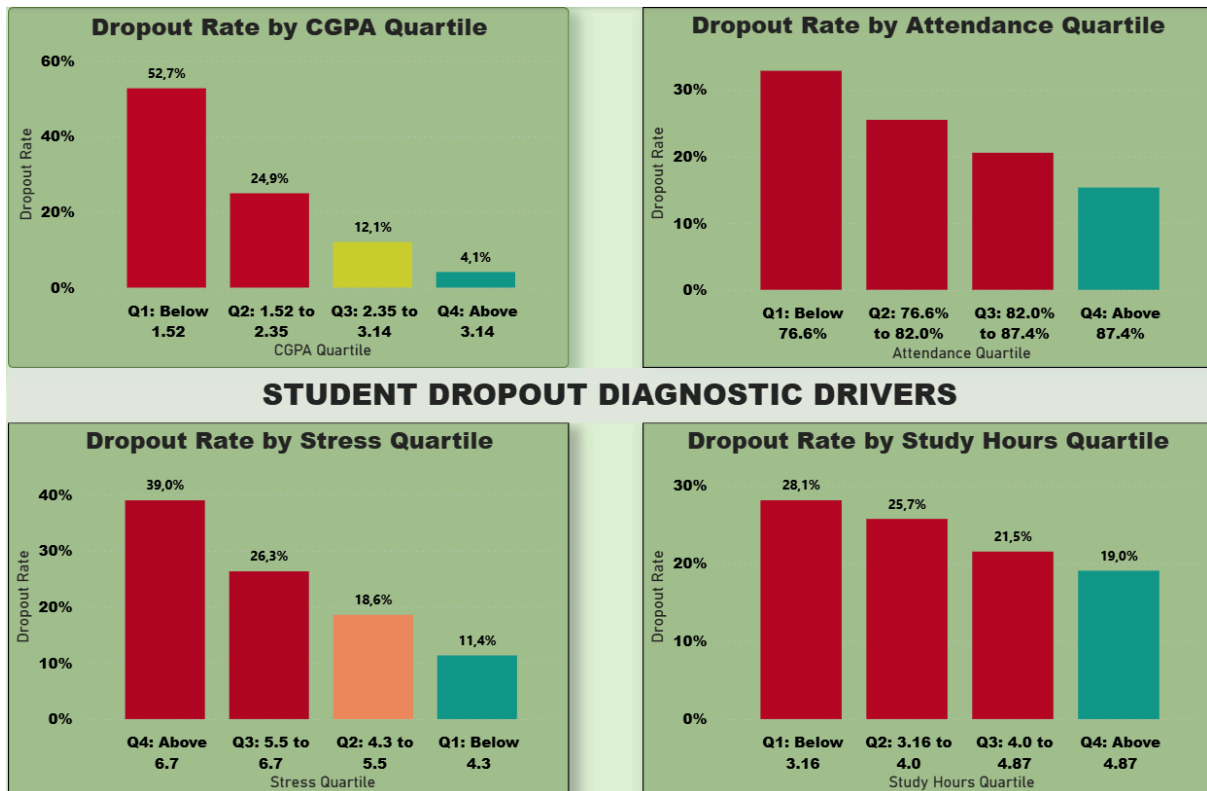
Recommendations

High Priority:

- Launch an early-warning CGPA alert system — auto-flag students below ~1.5 CGPA or a full-band drop, routed to advisors within 2 weeks.
- Attendance-triggered outreach — automatic advisor check-in when attendance falls below 80%.
- Expand mental health & stress-management support — proactive counseling and stress screening each semester.

Medium Priority:

- Study-skills & time-management coaching for students studying under 3 hours/day.
- Assignment-delay monitoring — treat 3+ day repeated lateness as an early behavioral flag.
- Close the digital-access and work-life gap with subsidized internet/devices and flexible scheduling.



Bar Color Legend (Meaning)

■ Red Bar.

Critical student dropout risk – Poor performance, immediate intervention required.

■ Orange Bar

Moderate student dropout risk – Below target, requires prompt support and close monitoring.

■ Yellow Bar

Low student dropout risk – Fair performance, some concerns, needs improvement and periodic monitoring.

■ Green Bar

Minimal or no student dropout risk – Good performance, on target, progressing well

5. Intervention Guide Summary

A separate advisor-facing Intervention Guide was produced to accompany the dashboard.

The guide provides:

- Plain-language explanations for each risk tier.
- Common warning indicators.
- Practical, non-punitive advisor actions.
- Suggested follow-up strategies appropriate for each level of risk.

The objective is to encourage supportive engagement rather than punitive action, promoting student success through timely intervention.

- **Support students with low academic performance** because they are the most likely to drop out.
- **Monitor attendance regularly** and assist students whose attendance starts to decline.
- **Provide stress management and counselling services** to help students cope with academic pressure.
- **Encourage consistent study habits** by promoting regular daily study time.
- **Follow up on late assignments early** to prevent students from falling behind.
- **Improve internet access and support students with part-time jobs**, as these factors can affect their studies.
- **Focus on student behavior and academic progress rather than demographics**, since performance and engagement have a much bigger impact on dropout than background.

6 Tech Stack & AI Usage

Tech Stack

- **Microsoft Excel** – Data cleaning, preprocessing, and exploratory data analysis (EDA)
- **SQL** – Data querying and analytical insights
- **Python** – Data analysis, visualization, and EDA

- **Power BI** – Interactive dashboard development and KPI reporting
- **Git & GitHub** – Version control, repository management, and project collaboration

AI Assistance (ChatGPT & Claude)

- Guided data cleaning and handling of missing values
- Supported EDA and analytical approach selection
- Generated and explained SQL queries and Python code
- Recommended Power BI dashboard design, KPIs, and DAX measures
- Assisted with Git/GitHub troubleshooting and project documentation
- **All AI-generated outputs were independently reviewed, validated, and adapted before inclusion in the final project.**

7. Limitations & Next Steps

This project was developed as a proof of concept and therefore does not include:

- Integration with a live Student Information System (SIS).
- Real-time data pipelines or automated data refresh.
- Direct counselling or intervention delivery.
- Long-term model monitoring or retraining.
- Findings are specific to the available dataset and may not generalize to all institutions.

Future work could improve the system by incorporating additional variables such as counselling participation, extracurricular involvement, semester-by-semester academic progression, and other institutional support indicators.

7.1 Risk Management Summary

Risk management was integrated throughout the project lifecycle to support successful delivery within the one-week timeline. Key risks identified included data quality issues, project schedule constraints, model

performance challenges, dashboard integration, team availability, and documentation delays.

To reduce these risks, the team conducted early data profiling, maintained regular progress meetings, prioritized critical tasks, used shared documentation and version control, and continuously monitored milestone completion. These mitigation strategies helped minimize disruptions and ensured the project remained on schedule.

7.2 Lessons Learned

Several valuable lessons emerged during the execution of this project:

- Effective planning and clearly defined responsibilities improved coordination across the team.
- Regular communication helped identify challenges early and reduced project delays.
- Close collaboration between Technical Project Managers and Data Analysts ensured that project objectives remained aligned with technical implementation.
- Transparent documentation of AI-assisted work promoted accountability and reproducibility.
- Working within a one-week sprint emphasized the importance of prioritization, adaptability, and continuous progress monitoring.

8. Conclusion

This project demonstrates how educational data can be transformed into actionable insights that support student success. By combining exploratory data analysis, predictive modeling, and an intuitive risk dashboard, the proposed Early Warning System provides institutions with a practical tool for identifying students who may require timely academic intervention.

Although developed as a bootcamp proof of concept, the solution illustrates the value of data-driven decision-making in improving student retention,

promoting equitable educational outcomes, and supporting institutional planning. With additional data and deployment within institutional systems, the project has the potential to evolve into a practical decision-support tool for higher education.

9.

10. Acknowledgements

The team expresses its sincere appreciation to the Women Techsters Bootcamp facilitators and mentors for their guidance, support, and constructive feedback throughout the project.

We also acknowledge the responsible use of AI-assisted tools, including ChatGPT and Claude, which supported documentation, coding assistance, analytical guidance, and report drafting. All AI-generated outputs were reviewed, validated, and refined by team members before inclusion in the final deliverables.

11. AI-Usage Ledger

The table below logs all use of generative AI tools (ChatGPT/Claude) throughout the project, per the team's AI Disclosure Drafting task. Entries should be added continuously as work is completed.

Date	Team Member	Tool Used	Purpose / Task	AI-Generated vs. Human-Reviewed
[date]	[name]	[ChatGPT / Claude]	[e.g., drafted data-cleaning script]	[AI-drafted, human-reviewed and tested]
[date]	[name]	[ChatGPT / Claude]	[e.g., EDA visualization code]	[AI-drafted, human-modified]
[date]	[name]	[ChatGPT / Claude]	[e.g., report section drafting]	[AI-drafted, human-edited]

Note: Replace placeholder rows with the team's actual logged entries from Day 4 onward, including any AI assistance used for the dashboard build or this report itself.